COMPONENT SIDE — USB at left

70 × 50 mm • 24 × 18 holes • 2.54 mm pitch • all point-to-point wires are on rear.

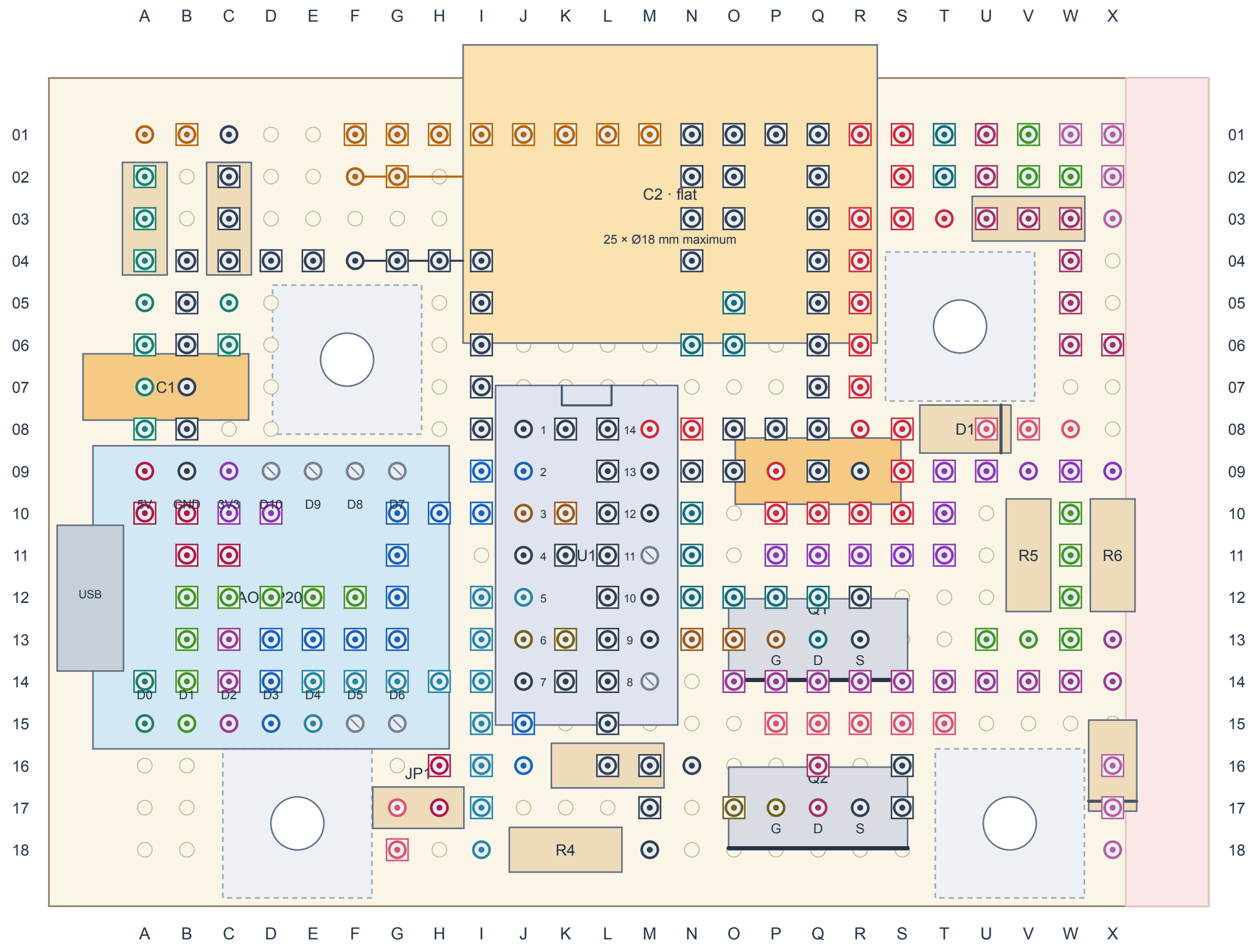
Solid dots: through-hole leads. Square ports: rear-only cable attachment pads.

Coordinates are assigned by this drawing; supplier lettering may be on the opposite face.

R1 100k · R2 22k · R3/R4 10k · R5/R6 4.7k · R7 3.3k

C1/C3 0.1 µF · D1/D3 NTE585 · JP1 removable shunt (remove for USB)

D2 stays at the pump; C4 stays at the Hall sensor. Neither mounts on this board.



U1 notch ↑ • Q1/Q2 printed faces ↓ • D1 stripe → W08 • D3 stripe → X18 • C2 + F02 / − F04

Grey squares: 9 × 9 mm mounting keep-outs. Centres (x,y): (18,17), (55,15), (15,45), (58,45) mm.

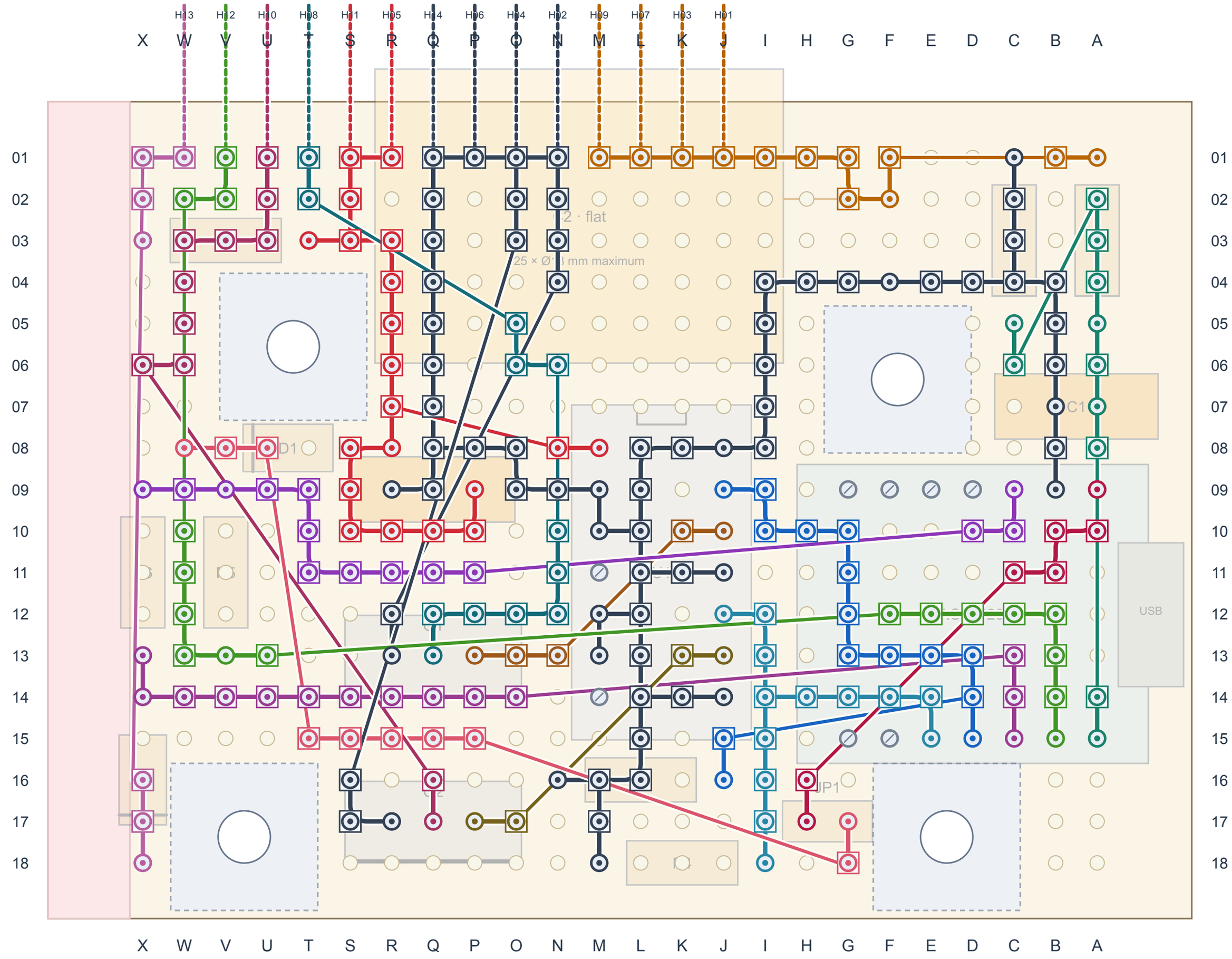
Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

SOLDER SIDE — mirrored

Flip the board left ↔ right; top stays top. Pad names do not change.

Insulated wires cross without joining. Solder at marked anchors; crossings do not join.

Bare tracks are anchored at each bend and at most three holes apart.



Solder-side drawing shows every wire; use the wire schedule or interactive net filter to trace crowded areas.

Grey squares: 9 × 9 mm mounting keep-outs. Centres (x,y): (18,17), (55,15), (15,45), (58,45) mm.

Grid offsets are nominal. Verify the board and dry-fit C2 before drilling or soldering. This is not a 1:1 drill template.

CABLE CONNECTIONS AND COLOUR KEY

Dashed leads attach to rear copper pads. Last column: wire centre behind the board.

H01	J01	S1 switched output	1 mm
H02	N01	Battery negative	1 mm
H03	K01	Buck IN+	1 mm
H04	O01	Buck IN−	1 mm
H05	R01	Buck OUT+ (set to 5.0 V)	1 mm
H06	P01	Buck OUT−	1 mm
H07	L01	Pump + / D2 striped end	1 mm
H08	T01	Pump − / D2 unstriped end	1 mm
H09	M01	LED harness +	1 mm
H10	U01	LED harness −	1 mm
H11	S01	Lock +5 V: A3144 pin 1 / C4	1 mm
H12	V01	Lock HALL: A3144 pin 3	1 mm
H13	W01	Lock SW: microswitch NO	1 mm
H14	Q01	Lock GND: Hall pin 2 / COM / C4	1 mm

OFF-BOARD CONNECTIONS

Battery + → S1 input. S1 output → H01. Battery − → H02. (No charge jack: charge at the splitter's female end.)
Set the buck output to 5.0 V before connecting H05/H06.
D2 NTE586 across pump terminals: stripe to pump +. C4 0.1 µF across A3144 pins 1–2.
A3144: printed face toward you, leads down; pins 1/2/3 = +5V/GND/HALL.
Twist switch COM → GND, NO → SW; NC unused. Lock cable uses the JST PH pair.
JP1: fit shunt for battery operation; remove for USB and follow build-guide power procedure.

- Switched battery
- Ground
- Regulated 5 V
- D1 → JP1
- JP1 → XIAO 5V
- 3.3 V
- Battery sense
- Hall signal
- Twist GPIO
- Switch contact
- Pump PWM
- LED PWM
- Pump gate
- LED gate
- Pump return
- LED return

Cable H02 attaches to N01. Capacitor C2 positive uses F02.

WIRE AND SOLDER ANCHORS

Solder every listed hole. Bare tracks are supported at bends and ≤3 holes apart.

W01	C13 → C15	TWIST	BARE track
W02	D14 → D15	PWM_P	BARE track
W03	I14 → F14 → E14 → E15	PWM_L	BARE track
W04	X16 → X18	SW	BARE track
W05	T02 → T01	DRAIN_P	BARE track
W06	X06 → W06 → W03 → U03 → U01	DRAIN_L	BARE track
W07	W02 → V02 → V01	HALL	BARE track
W08	N01 → N04	GND	BARE track
W09	O01 → O03	GND	BARE track
W10	Q16 → Q17	DRAIN_L	BARE track
W11	N01 → O01	GND	BARE track
W12	O01 → P01	GND	BARE track
W13	P01 → Q01	GND	BARE track
W14	Q08 → Q09 → R09	GND	BARE track
W15	Q01 → Q04 → Q07 → Q08	GND	BARE track
W16	Q08 → O08 → O09 → M09	GND	BARE track
W17	M09 → M10	GND	BARE track
W18	L12 → M12	GND	BARE track
W19	L11 → L12	GND	BARE track
W20	L10 → L11	GND	BARE track
W21	M10 → L10	GND	BARE track
W22	M12 → M13	GND	BARE track
W23	L11 → J11	GND	BARE track
W24	L14 → J14	GND	BARE track
W25	L12 → L14	GND	BARE track
W26	M16 → N16	GND	BARE track
W27	L14 → L16 → M16	GND	BARE track
W28	M16 → M18	GND	BARE track
W29	L10 → L08 → J08	GND	BARE track
W30	J08 → I08 → I05 → I04 → F04	GND	BARE track
W31	C04 → C01	GND	BARE track
W32	F04 → C04	GND	BARE track

6 mm removable spacers + existing 10 mm posts. Printed parts stay unchanged.

WIRE AND SOLDER ANCHORS

Solder every listed hole. Bare tracks are supported at bends and ≤3 holes apart.

W33	C04 → B04 → B07	GND	BARE track
W34	B07 → B09	GND	BARE track
W35	S03 → S01	5V_REG	BARE track
W36	T03 → S03	5V_REG	BARE track
W37	S01 → R01	5V_REG	BARE track
W38	R07 → R08	5V_REG	BARE track
W39	S03 → R03 → R06 → R07	5V_REG	BARE track
W40	R08 → S08 → S10 → P10 → P09	5V_REG	BARE track
W41	N08 → M08	5V_REG	BARE track
W42	R12 → R13	GND	BARE track
W43	S16 → S17 → R17	GND	BARE track
W44	I14 → I12 → J12	PWM_L	BARE track
W45	I18 → I15 → I14	PWM_L	BARE track
W46	N13 → P13	GATE_P	BARE track
W47	J10 → K10	GATE_P	BARE track
W48	J16 → J15	PWM_P	BARE track
W49	D14 → D13 → G13 → G10 → I10 → I09 → J09	PWM_P	BARE track
W50	J01 → K01	VLOAD	BARE track
W51	K01 → L01	VLOAD	BARE track
W52	L01 → M01	VLOAD	BARE track
W53	J01 → G01 → G02 → F02	VLOAD	BARE track
W54	B01 → A01	VLOAD	BARE track
W55	F02 → F01	VLOAD	BARE track
W56	A05 → A07	ADC	BARE track
W57	A14 → A15	ADC	BARE track
W58	A07 → A08	ADC	BARE track
W59	C06 → C05	ADC	BARE track
W60	A05 → A02	ADC	BARE track
W61	P11 → S11 → T11 → T09 → V09	3V3	BARE track
W62	C09 → C10 → D10	3V3	BARE track
W63	V09 → X09	3V3	BARE track
W64	O14 → R14 → U14 → X14	TWIST	BARE track

6 mm removable spacers + existing 10 mm posts. Printed parts stay unchanged.

WIRE AND SOLDER ANCHORS

Solder every listed hole. Bare tracks are supported at bends and ≤3 holes apart.

W65	X14 → X13	TWIST	BARE track
W66	V13 → W13 → W10	HALL	BARE track
W67	F12 → C12 → B12 → B15	HALL	BARE track
W68	V13 → U13	HALL	BARE track
W69	X01 → W01	SW	BARE track
W70	X01 → X03	SW	BARE track
W71	C11 → B11 → B10 → A10 → A09	XIAO_5V	BARE track
W72	H17 → H16	XIAO_5V	BARE track
W73	O17 → P17	GATE_L	BARE track
W74	J13 → K13	GATE_L	BARE track
W75	N06 → O06 → O05	DRAIN_P	BARE track
W76	Q13 → Q12 → N12 → N10	DRAIN_P	BARE track
W77	U08 → W08	AFTER_D1	BARE track
W78	P15 → S15 → T15	AFTER_D1	BARE track
W79	G17 → G18	AFTER_D1	BARE track
W80	Q16 → X06	DRAIN_L	Rounded arch · top 6.9 mm
W81	N08 → R07	5V_REG	Rounded arch · top 6.9 mm
W82	R12 → N04	GND	Rounded arch · top 9.9 mm
W83	S16 → O03	GND	Rounded arch · top 12.9 mm
W84	N13 → K10	GATE_P	Rounded arch · top 6.9 mm
W85	D14 → J15	PWM_P	Rounded arch · top 6.9 mm
W86	B01 → F01	VLOAD	Rounded arch · top 6.9 mm
W87	A14 → A08	ADC	Rounded arch · top 6.9 mm
W88	C06 → A02	ADC	Rounded arch · top 6.9 mm
W89	P11 → D10	3V3	Rounded arch · top 9.9 mm
W90	O14 → C13	TWIST	Rounded arch · top 9.9 mm
W91	W02 → W10	HALL	Rounded arch · top 9.9 mm
W92	F12 → U13	HALL	Rounded arch · top 9.9 mm
W93	X01 → X16	SW	Rounded arch · top 12.9 mm
W94	C11 → H16	XIAO_5V	Rounded arch · top 12.9 mm
W95	O17 → K13	GATE_L	Rounded arch · top 6.9 mm
W96	T02 → O05	DRAIN_P	Rounded arch · top 6.9 mm

6 mm removable spacers + existing 10 mm posts. Printed parts stay unchanged.

WIRE AND SOLDER ANCHORS

Solder every listed hole. Bare tracks are supported at bends and ≤3 holes apart.

W97	N06 → N10	DRAIN_P	Rounded arch · top 9.9 mm
W98	U08 → T15	AFTER_D1	Rounded arch · top 12.9 mm
W99	P15 → G18	AFTER_D1	Rounded arch · top 9.9 mm

6 mm removable spacers + existing 10 mm posts. Printed parts stay unchanged.